

Thesis TITLE

Summary

Candidate:

name surname

Supervisors:

prof. name surname

prof. name surname

prof. name surname

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Argumentative Graph-RAG for Participatory Democracy

Michele Pagano Mariano

Abstract

Participatory democracy platforms generate massive volumes of textual data, leading to severe information overload for both citizens and decision-makers. While Natural Language Processing (NLP) and Argument Mining (AM) can extract argumentative structures, navigating this complex data remains a challenge.

This thesis explores the integration of Argument Mining with Retrieval-Augmented Generation (RAG) to create "Graph-RAG." By constructing knowledge graphs that represent the explicit relationships (support, attack, equivalence) between citizen proposals, we aim to enhance the ability of Large Language Models (LLMs) to retrieve context-rich information and generate balanced summaries.

We propose a pipeline that first builds argumentative graphs from civic datasets and then utilizes graph traversal to power a RAG system. This approach addresses the limitations of standard text-based retrieval by exploiting the structural logic of debates, ultimately providing intelligent tools for navigating large-scale deliberative discussions.